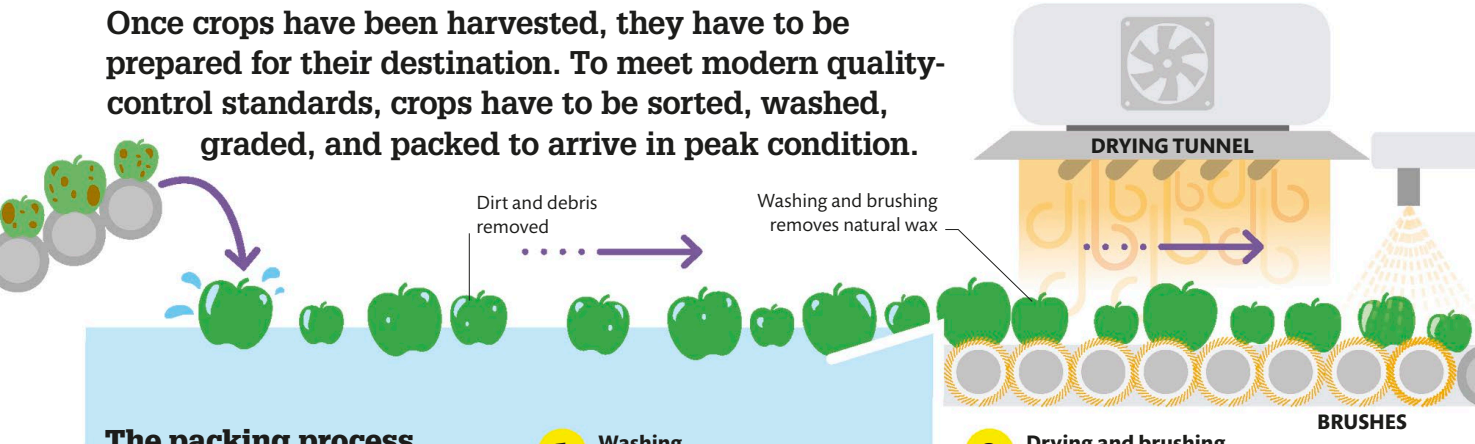
Sorting and packing

Once crops have been harvested, they have to be prepared for their destination. To meet modern quality-control standards, crops have to be sorted, washed, graded, and packed to arrive in peak condition.

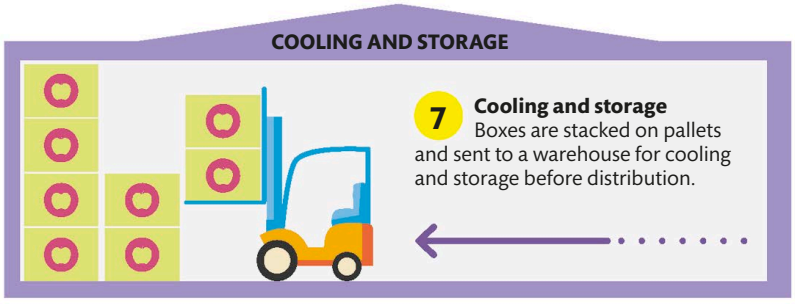


The packing process

Getting a fresh product to customers involves various cleaning, grading, and packing processes. Automation is replacing this once labor-intensive process with optical recognition systems and sorting devices. The process can be adapted to all kinds of fruits and vegetables, from heavy, muddy potatoes to delicate grapes.

1 Washing
Washing is done by submersion in tanks or by overhead sprayers. Mild detergent is added to help remove farm chemicals, pathogens, and dirt.

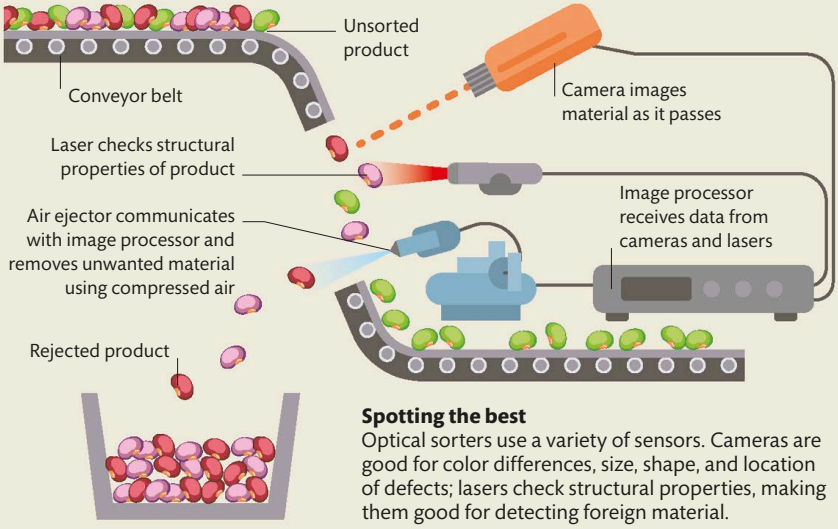
2 Drying and brushing
The crops are dried as they pass over rotating brushes, which dislodge any surface deposits that were not removed during the washing process.



7 Cooling and storage
Boxes are stacked on pallets and sent to a warehouse for cooling and storage before distribution.

Optical sorting

Packing houses often use optical sorters to process crops. The items pass over or under sensors, either on a conveyor or, in freefall sorters (right), while falling. The sensors are connected to an image-processing system. The passing objects are compared with predefined criteria for selection. Rejected items trigger the separation system: a blast of compressed air for small items or mechanical retrieval for larger produce. The rejected material is dumped, while the rest carries on for further processing.



Spotting the best
Optical sorters use a variety of sensors. Cameras are good for color differences, size, shape, and location of defects; lasers check structural properties, making them good for detecting foreign material.